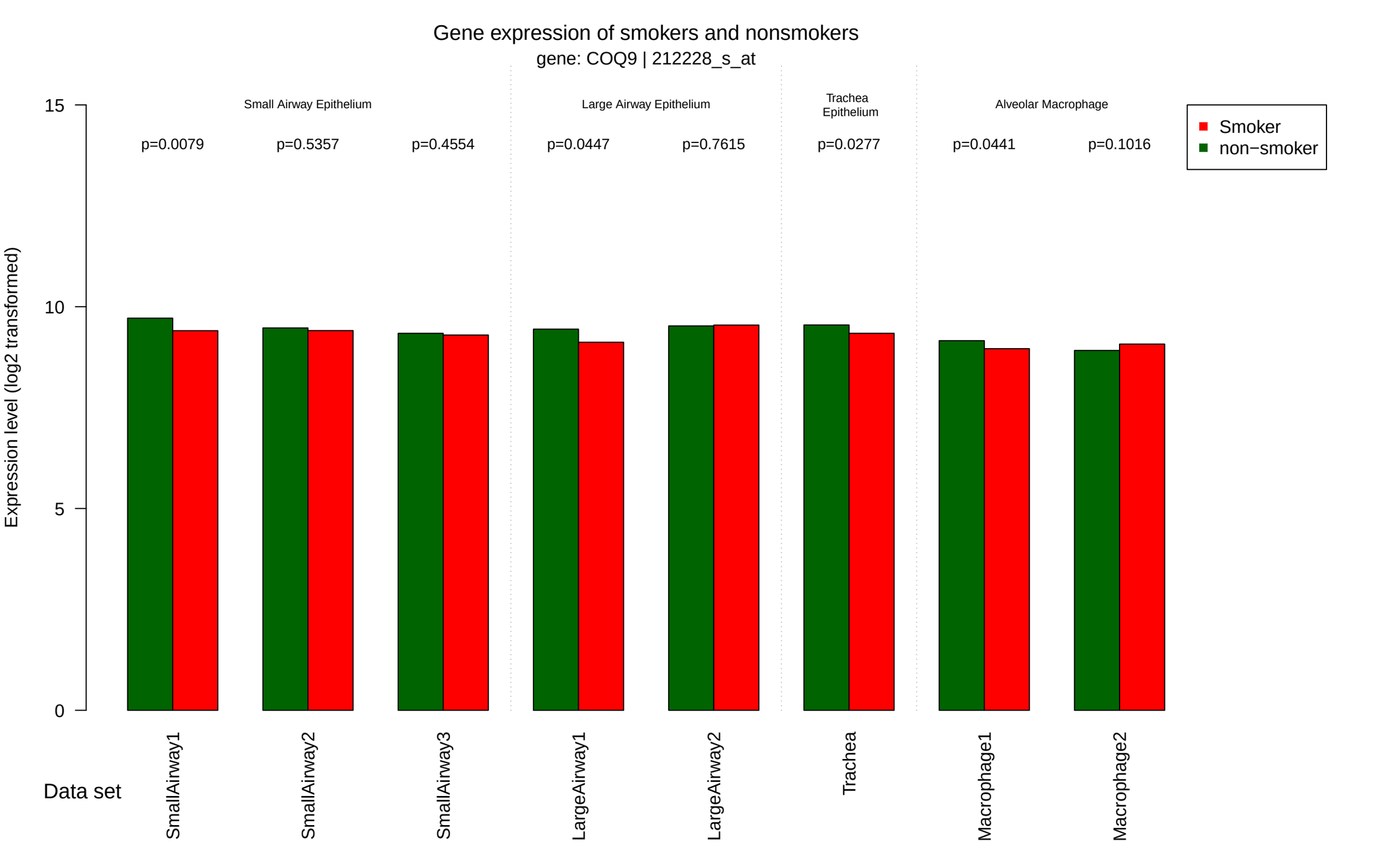
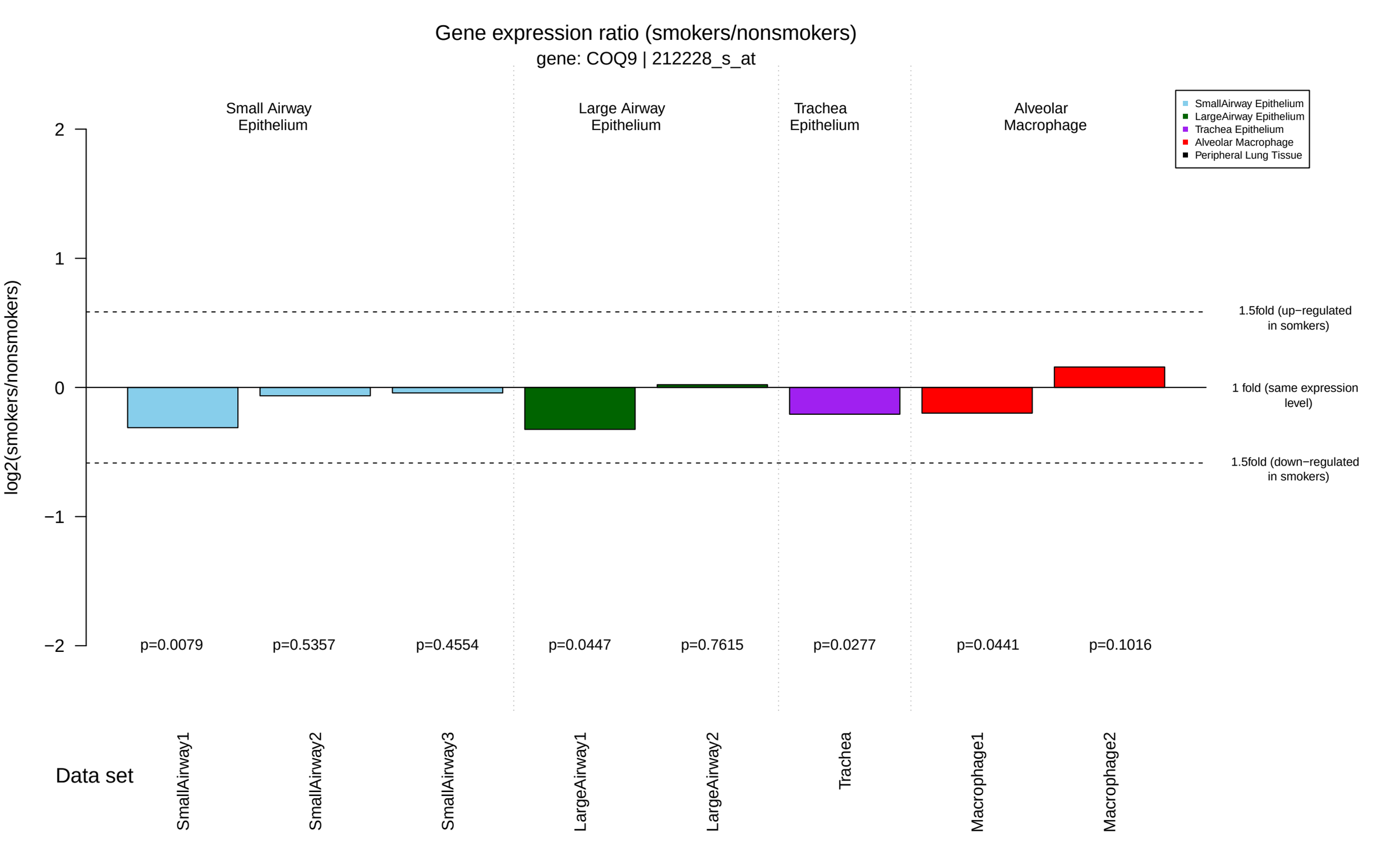
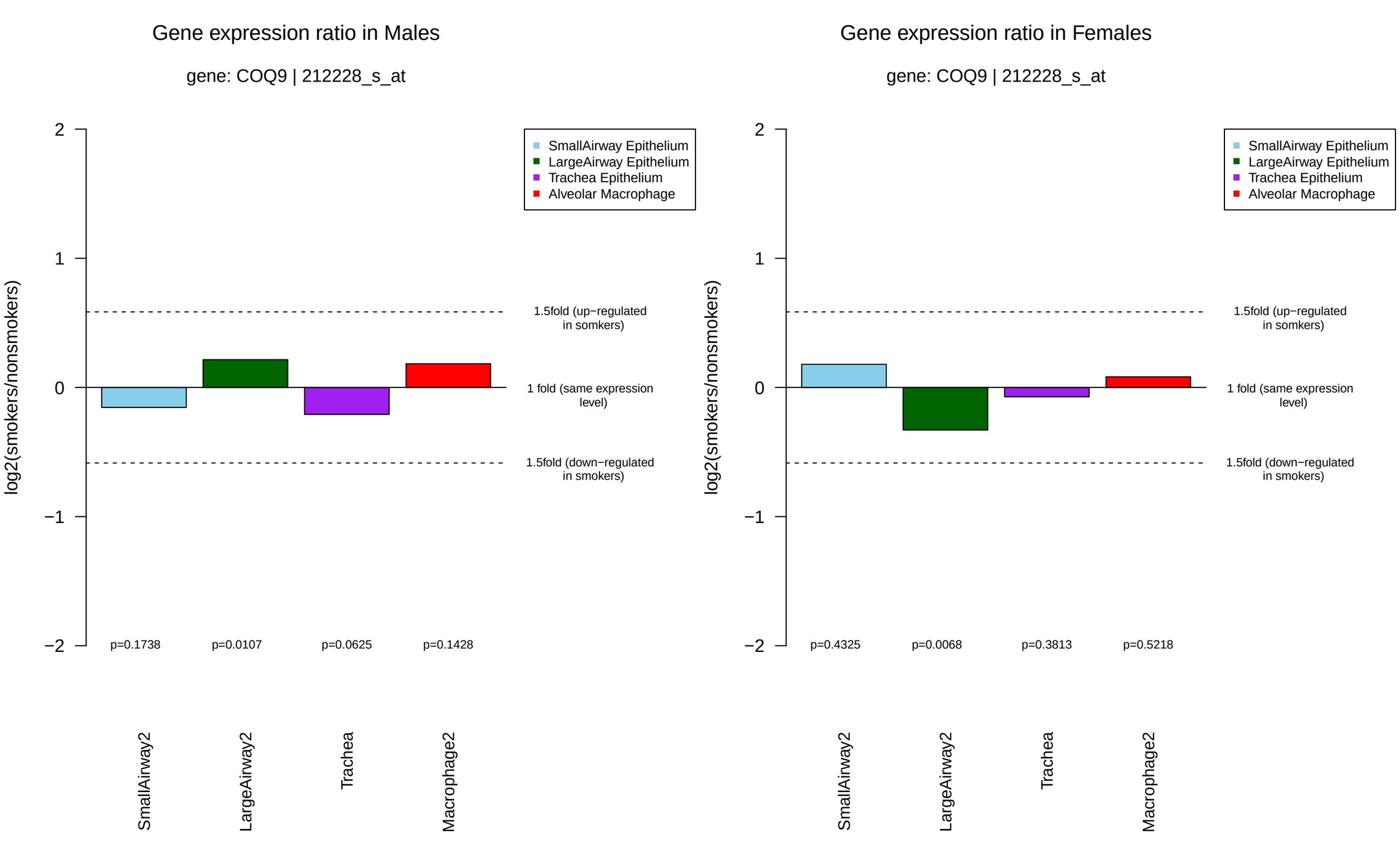
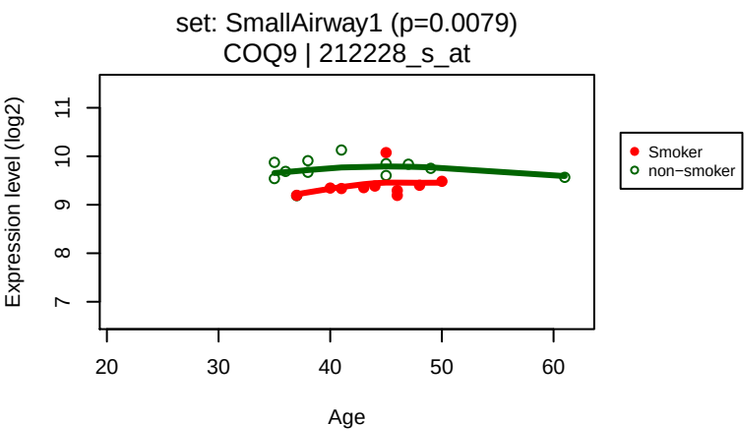
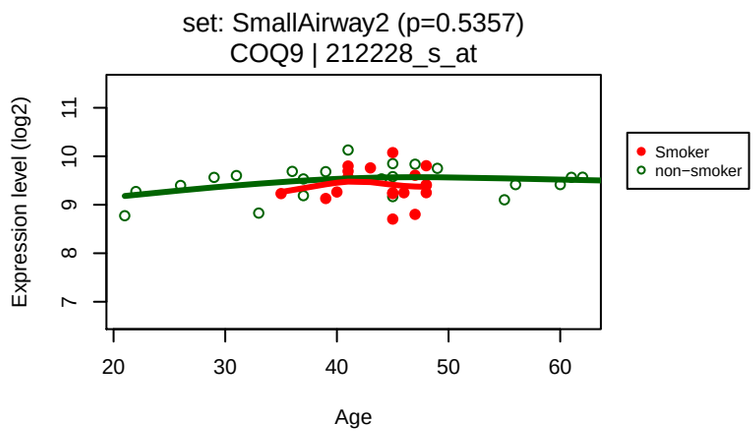
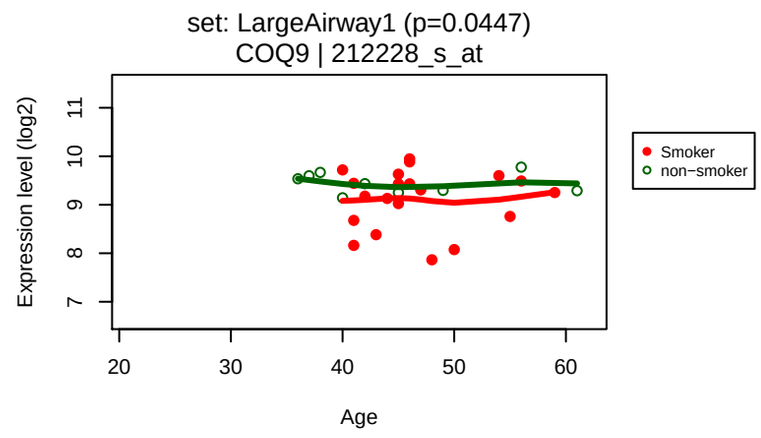

Gene expression of smokers and nonsmokers
gene: COQ9 | 212228_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0079
p=0.5357
p=0.4554
p=0.0447
p=0.7615
p=0.0277
p=0.0441
p=0.1016
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

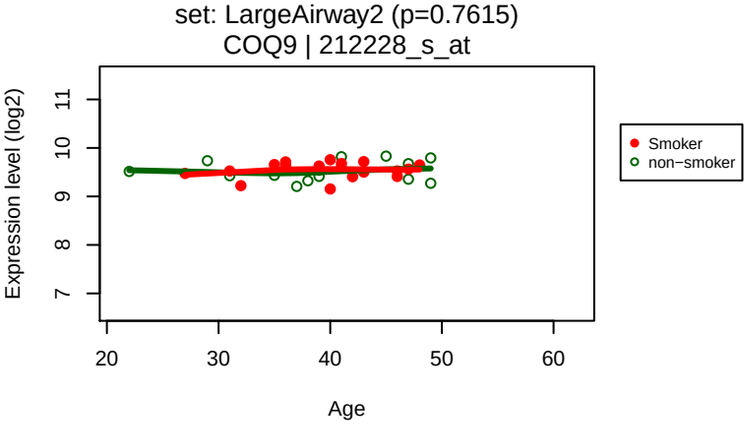

Gene expression ratio (smokers/nonsmokers)
gene: COQ9 | 212228_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0079
p=0.5357
p=0.4554
p=0.0447
p=0.7615
p=0.0277
p=0.0441
p=0.1016
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

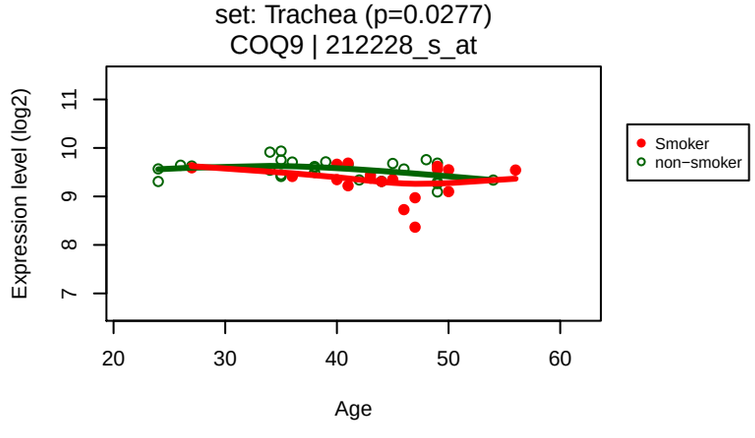

Gene expression ratio in Males
gene: COQ9 | 212228_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1738
p=0.0107
p=0.0625
p=0.1428
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COQ9 | 212228_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4325
p=0.0068
p=0.3813
p=0.5218
SmallAirway2
LargeAirway2
Trachea
Macrophage2

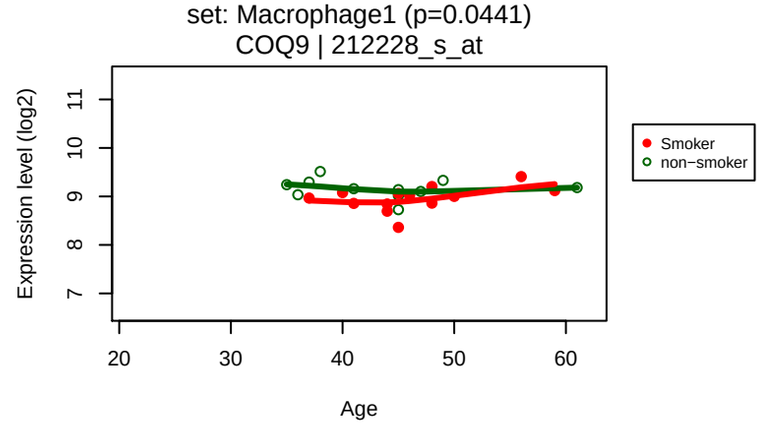

set: SmallAirway1 (p=0.0079)
COQ9 | 212228_s_at
Expression level (log2)
Age
Smoker
non−smoker

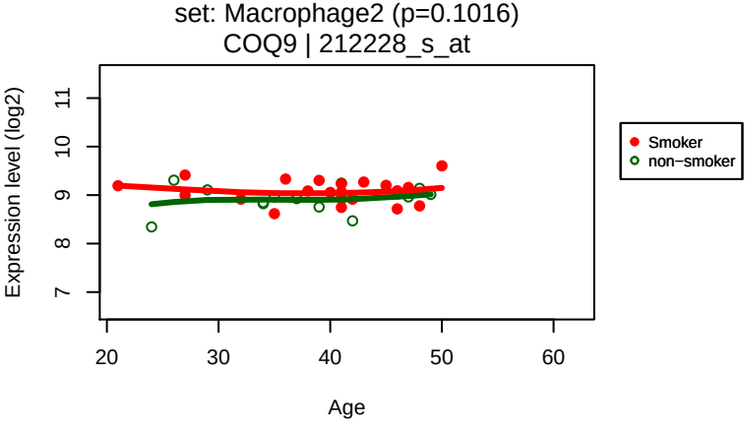

set: SmallAirway2 (p=0.5357)
COQ9 | 212228_s_at
Expression level (log2)
Age
Smoker
non−smoker

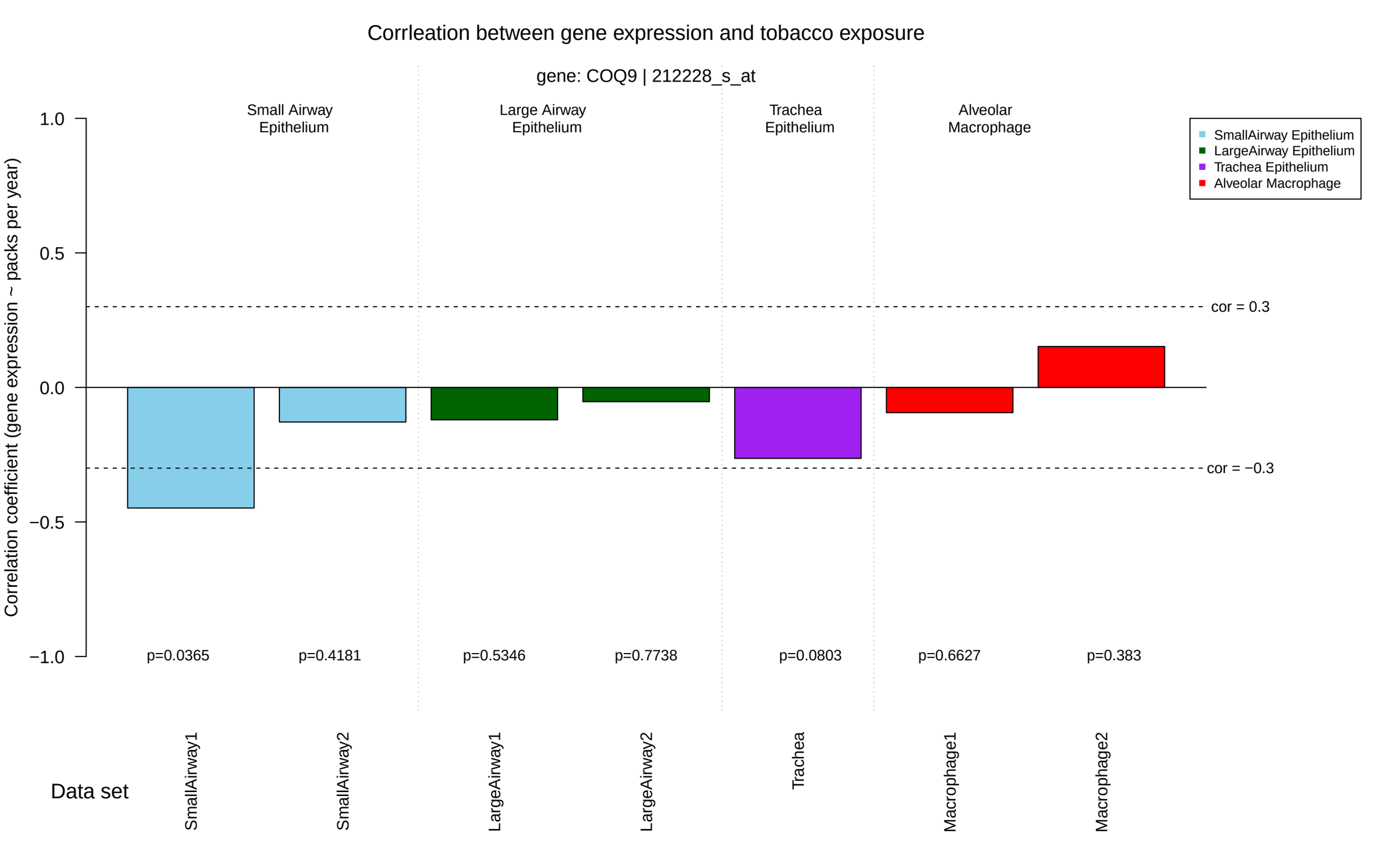

set: LargeAirway1 (p=0.0447)
COQ9 | 212228_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7615)
COQ9 | 212228_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0277)
COQ9 | 212228_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0441)
COQ9 | 212228_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1016)
COQ9 | 212228_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COQ9 | 212228_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0365
p=0.4181
p=0.5346
p=0.7738
p=0.0803
p=0.6627
p=0.383
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2